COHERENCE OF TOR SHEAVES AND DERIVED TENSOR PRODUCTS ON LOCALLY NOETHERIAN SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 08DX: sources/stacks-chow/tags.txt, source lines 10653–10653. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0AZ4: sources/stacks-chow/tags.txt, source lines 13812–13812. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 08E8: sources/stacks-chow/tags.txt, source lines 10664–10664. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09J3: sources/stacks-chow/tags.txt, source lines 12061–12061. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a locally Noetherian scheme.*

(1) *If $\mathcal{F}$ and $\mathcal{G}$ are coherent $\mathcal{O}_X$-modules, then $Tor_p^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is too.*
(2) *If L and K are in $D^-_{Coh}(\mathcal{O}_X)$, then so is $L \otimes^{\mathbf{L}}_{\mathcal{O}_X} K$.*

Proof. Let us explain how to prove (1) in a more elementary way and part (2) using previously developed general theory.

Proof of (1). Since formation of Tor commutes with localization we may assume X is affine. Hence $X = \mathrm{Spec}(A)$ for some Noetherian ring A and $\mathcal{F}$, $\mathcal{G}$ correspond to finite A-modules M and N (Cohomology of Schemes, Lemma 01XZ). By Derived Categories of Schemes, Lemma 08DX we may compute the Tor's by first computing the Tor's of M and N over A, and then taking the associated $\mathcal{O}_X$-module. Since the modules $\mathrm{Tor}_p^A(M, N)$ are finite by Algebra, Lemma 0AZ4 we conclude.

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By Derived Categories of Schemes, Lemma 08E8 the assumption is equivalent to asking L and K to be (locally) pseudo-coherent. Then $L \otimes^{\mathbf{L}}_{\mathcal{O}_X} K$ is pseudo-coherent by Cohomology, Lemma 09J3. $\square$